

Low-Noise Transimpedance Amplifier Design Procedure for Optical Communications

Shahab Shahdoost, and Bardia
Bozorgzadeh
EECS Dept. of Case Western Reserve
University

Ali Medi
EE Dept. of
Sharif University of Technology

Namdar Saniei
ECSE Dept. of University of Ontario
Institute of Technology

Abstract— Here, we report on design and measurement results of a state of the art low-noise and high-gain transimpedance amplifier (TIA) implemented in 0.18 μm TSMC CMOS technology. In depth design methodology for design of high gain and low noise TIA for 2.5 Gb/s optical communication family is presented. A novel noiseless capacitive feedback is proposed and implemented as a noise efficient feedback for TIA circuits. Besides, analytical noise calculations in this family of TIA circuits are presented and optimum noise criteria are derived. The saturation and instability problem of TIA circuits resulted from DC dark current of the input photodiodes (PDs) is addressed and a circuit-level solution is proposed. The measurement results of 0.18 μm chip shows bandwidth of 52 kHz to 1.62 GHz, and transimpedance gain of 75.5 dB Ω while dissipating 26.3 mW from a 2.2 V power supply, including the output buffer. Taking advantage of proposed capacitive feedback network and optimum noise criteria, noise measurement results show average input referred current noise of 3.18 pA/ $\sqrt{\text{Hz}}$ for this TIA in the bandwidth of operation.

Keywords—Current-input circuits, low-noise circuits, optical communications, transimpedance amplifiers

I. INTRODUCTION

Recently, transimpedance amplifiers are being used in a very wide range of applications such as optical communications (families of 2.5, 10, 25,... Gb/s), mechanical sensors, biosensors, and other specific bio-applications like DNA sequencing, and impedance spectroscopy [1-4]. This broad range of applications results in different sets of constraints for a circuit designer when designing a TIA. While in high speed families of optical communication circuits the main goal is pushing the bandwidth of TIA higher in GHz range [4], bio-application TIAs usually stay in much lower frequency ranges but at the same time demand better noise performances and lower power consumption for chronic/long-term implantable versions [1-3]. So, the first question to be addressed in a TIA design is which application the TIA is being used for and what are the specific design challenges for that specific application.

Generally, as the first building block in a receiving front end, no matter what the application is, TIAs convert the input current to an amplified voltage for use in the next blocks. System performance in terms of bandwidth, sensitivity and

signal to noise ratio (SNR) is strongly influenced by its first building block. Hence, having enough bandwidth for the application, high and precise transimpedance gain in the bandwidth of interest, and low input noise current for higher sensitivity while dissipating reasonable power are the fundamental features expected from a well-designed TIA.

In TIA design for 2.5 Gb/s optical communications family, the fact that the bandwidth is still in range of few GHz, leads to higher gain and lower noise, at the same time, to be the most important design challenges. Increasing gain usually increases chance of instability and low noise performance usually translates into higher power consumption. Especially in terms of instability, in addition to typical issues that designers should take into account for a voltage amplifier, such as phase margin and gain margin, saturation and instability resulting from DC component of the input current can add an extra level of difficulty for TIA design [4]. The input current is not ideally a symmetric AC current at the input of TIA, and usually there is a non-ideal DC current component. In optical communication circuits, this DC current comes from mismatch between the two series inverse-biased input photodiodes that are used to sense the optical beams at the end of optical fiber. For each of the mentioned problems, understanding the problem and fundamental limitations is the key issue which can result in finding an optimum solution.

General accepted topology for TIA circuit is a voltage amplifier with shunt-shunt feedback across it, as shown in Fig. 1. This feedback topology is opted because of low input impedance, as desired for a current-input block and low output impedance, as needed for a voltage-output amplifier. This feedback network also assures almost constant transimpedance gain in the bandwidth of interest, while decreasing sensitivity to process and temperature variation.

For the voltage amplifier, designer has plenty of choices. Because the noise performance of a TIA is another important design challenge, the simpler voltage amplifier with less number of active components is opted, the less input current-noise would be resulted. So, a single-stage amplifier can be a viable candidate if enough gain can be achieved. Choice of input topology for voltage amplifier is mostly dictated by noise and bandwidth constraints.

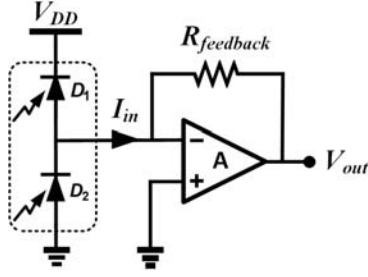


Fig. 1. Typical TIA topology; a voltage amplifier with shunt resistive feedback.

Input parasitic capacitances of photodiode, electro-static discharge (ESD) circuit, and input PAD at the input of TIAs primarily constrain their bandwidth and noise performances [6]. As a result, different input topologies have been proposed to relax the effect of input capacitance, such as common gate (CG) [7], regulated cascode (RGC) [6,17], common drain (CD) [8], and common source (CS) [5]. At first glance, CG and RGC topologies seem better candidates in terms of relaxing the effect of input parasitic capacitances, but high current noise contribution of the load and input transistors in these topologies, degrades the noise performance drastically [4]. Isolating property of CD topology can be useful but it cannot be a good choice, first because we still need another stage for amplification, and also high noise contribution of the buffer transistor in addition to amplified (divided by smaller than unity gain of buffer) noise from second stage makes this topology a poor candidate in terms of noise efficiency. So, a voltage amplifier with a CS topology in its input would be the first step in the design procedure.

For choice of feedback network, the most common approach is a resistive feedback network, as shown in Fig. 1. Resistor value doesn't change in the bandwidth of interest and also provides DC current path for the dark current of PDs, although in most cases because of headroom problems another DC path is designed and implemented and this resistor doesn't serve as a DC path for input current [4]. However, the main drawback of resistive feedback is its noise performance. Current noise of this resistor is directly added to the input referred current noise of the TIA which degrades the noise performance drastically [4, 14]. A capacitive, noiseless feedback network can serve as a viable solution for this problem. As shown in Fig. 2, we proposed [5] a TIA with capacitive feedback network for maintaining high gain and low noise at the same time. Also a circuit level solution for the DC dark current in this topology is provided.

The rest of the paper is organized as follows; in section II, the basic TIA topology, analytical calculations of the TIA specifications including gain, band-width, and input referred noise are reviewed. Besides, noise analysis is performed and optimum criterion for noise calculation is derived. Also, a novel design procedure is proposed for this TIA family. In section III, circuit structures are shown and the final proposed

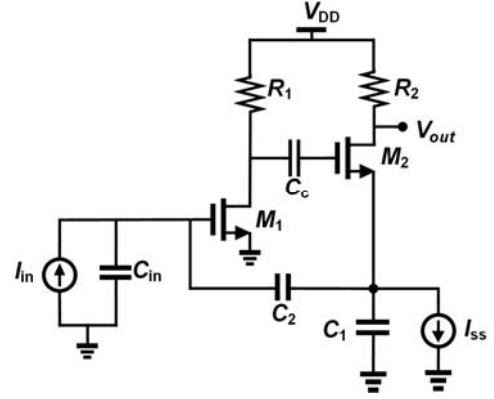


Fig. 2. Basic TIA topology with common source as input stage and capacitive feedback network.

topology is presented. In section IV, the measurement results of implemented TIA in 0.18 μm CMOS technology is and compared with other recent publications. Finally, section V. draws some conclusions from this work.

II. CHARACTERISTICS' FUNDAMENTALS

In this section, we go through the transimpedance gain, bandwidth, and input noise current characteristics of the proposed transimpedance amplifier with capacitive feedback which employs a common source stage as the core voltage amplifier. A simplified concept of such a TIA is shown in Fig. 2. In [5] we have shown that the -3dB bandwidth of the circuit can be calculated from the following equation.

$$f_{-3dB} \approx \frac{g_{m2} \cdot (1 + g_{m1}R_1)}{2\pi \cdot C_1} \quad (1)$$

in which, g_{m1} and g_{m2} are the transconductance of transistor M_1 and M_2 respectively. Basically Eq. 1 emphasizes that by increasing current consumption in each transistor, bandwidth of the circuit will be increased.

Transimpedance gain of the circuit in Fig. 2. can be calculated from Eq. 2 [5]

$$TI \text{ Gain} = \frac{V_{out}}{I_{in}} = \frac{R_2 \frac{(g_{m1}R_1(C_1 + C_2) + C_2)}{(C_{in} + C_2 + g_{m1}R_1C_2)}}{1 + s \frac{(C_1C_{in} + C_1C_2 + C_2C_{in})}{g_{m2}(C_{in} + C_2 + g_{m1}R_1C_2)}} \quad (2)$$

in which C_{in} is

$$C_{in} = C_{PD} + C_{gs} + (1 + g_{m1}R_1)C_{gd1} \quad (3)$$

The transimpedance gain for low frequencies can be expressed by $[R_2 \times (1 + C_1/C_2)]$.

The input referred current noise of the circuit, as thoroughly explained in [5], can be written as

$$\overline{I_{n,in}^2} = \frac{4KT\gamma}{g_{m1}} \cdot \left(1 + \frac{1}{g_{m1}R_1}\right) \cdot \left[\left(\frac{C_1}{1 + C_1/C_2} + C_{in}\right) \cdot s\right]^2 \quad (4)$$

in which K is the Boltzmann constant and γ represents the noise excess factor for the transistors.

To minimize this input noise current in the first step, we choose R_1 to be its maximum value with regard to voltage headroom limitations.

Also with regard to the relation between g_m of the transistor with its gate source capacitance and maximum cut-off frequency of the transistor, i.e. $g_m = 2\pi \cdot (f_T \times C_{gs})$, the optimum noise criterion can be derived as [5];

$$C_{gs} = C_{PD} + C_{ESD} + \frac{C_1}{1 + C_1/C_2} \quad (5)$$

A design guideline can be drawn from the derived equations for transimpedance gain, bandwidth, input noise current, and the criterion for optimized noise performance. C_1 can be calculated from the maximum required bandwidth by using Eq. 1, as long as power budget of the second stage has set a limit on g_{m2} . Transimpedance gain requirements can set the values of R_2 and C_2 considering Eq. 2. And finally, Eq. 5 dictates the aspect ratio for transistor M_1 . In this design $R_2 = 570 \Omega$ and $C_1 = 0.5 \text{ pF}$ were chosen.

III. FINAL CIRCUIT TOPOLOGY

In the previous section, fundamental characteristics of a capacitive feedback transimpedance amplifier in terms of transimpedance gain, noise, and bandwidth were explained. In this section, first the transimpedance amplifier performance characteristics are further improved by adding circuit level elements to the circuit without fundamentally changing the topology, and further ahead, a novel circuit level solution is suggested and explained for generally-known DC dark current problem in transimpedance amplifiers design.

In order to further decrease the input noise current contribution of R_1 and at the same time increasing the gain of the voltage amplifier, a PMOS transistor is added to the drain of M_1 and is biased optimally for minimum noise current. Also another NMOS transistor is stacked on top of transistor M_1 to decrease the Miller effect of its gate drain capacitance, as shown in Fig. 3.a.

In most cases, amplifiers which voltage are their output, needs to be matched to 50 ohm loads. To do so, another stage (M_{Buffer}) is added to the proposed transimpedance amplifier [5]. Besides, to satisfy the low cut-off frequency requirement of 50 KHz in optical communication standards, three-transistor topology depicted in Fig. 3.b. is implemented which plays the role of a very large resistor, while occupying considerably less silicon area compared to using a regular resistors on the chip with the same resistor value.

To address the dark-DC current, that leads to instability and saturation problems in transimpedance amplifier design, instead of using off-chip solutions [9], we proposed an efficient circuit level solution [5]. As shown in Fig. 3.c, a pair of NMOS and PMOS transistors has been implemented at the gate of input transistor. As described thoroughly in [5], this pair remains off while there is not DC current at the input of

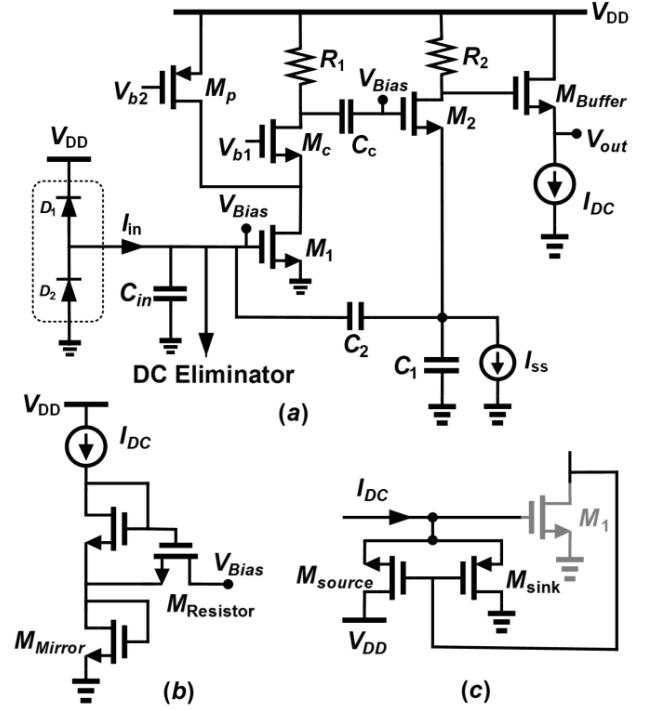


Fig. 3. (a) Main Circuit (b) Bias Network of M_1 and M_2 (c) Proposed DC Eliminator Circuit.

the transimpedance amplifier. In case of positive current at the input, M_{sink} is activated and sinks the extra current. Also, in presence of negative dark current, M_{source} will provide the current to the input node.

These NMOS and PMOS transistors are optimally designed with minimum widths to have negligible effect on the input noise current of the circuit. Besides that, the activation time period of this pair is very short, which in average has negligible noise contribution to the overall input noise current of the circuit.

To show the efficacy of the proposed DC-eliminator circuit, a 200 μA DC current, changed the DC bias point of the input transistor less than 6% in the post-layout simulation of the chip.

IV. MEASUREMENT RESULTS

Based on the design methodology described in previous sections, a transimpedance amplifier is fabricated in 0.18 μm TSMC CMOS technology. Fig. 4 shows the chip microphotograph of the implemented TIA which occupies $400 \times 300 \mu m^2$ of silicon die area.

In order to measure specifications of a TIA, different approaches can be taken. A common approach is measuring the S-Parameters of the circuit, just like a LNA, and then calculating the TIA's parameters by modified formulas. This approach is widely used because almost all the measurement set-ups are made for S-Parameters measurement and new instruments would not be needed. The measured S-Parameters of the circuit are shown in Fig. 5.

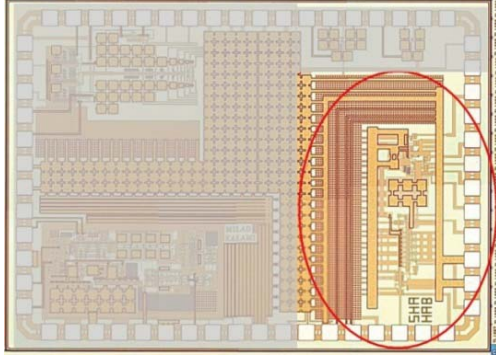


Fig. 4. The chip micrograph of implemented TIA in 0.18 μm technology

In order to calculate the transimpedance gain of a TIA from the shown S-parameters, the formula in Eq. 6 can be used [10].

$$Z_t = \frac{2 \cdot S_{21}}{(1 - S_{11}) \cdot (1 - S_{22}) - S_{21} S_{12}} \cdot Z_0 \quad (6)$$

This formula can be used only when the input of the TIA is directly connected to the network analyzer, and there is no input matching. By using the above formula and measured S-parameters, the transimpedance gain of the circuit is depicted in Fig. 6 and is compared with the simulation results.

The post-layout simulated transimpedance gain was a flat gain of 78 dB Ω with -3dB bandwidth between 52 kHz to 1.81 GHz, while as it can be seen in Fig. 6, the transimpedance gain in the measurement has been decreased less than 3 dB and was measured to be 75.5 dB Ω . Also, bandwidth of the circuit has decreased to 1.62 GHz. The peaking in the frequency response is less than 1.3 dB.

For noise measurement, further considerations must be taken into account. As the noise performance of the circuit can be degraded drastically by the ambient noise, a shielding Faraday cage must be used and the circuit must be installed properly in it. As the input must be opened while measuring the output voltage noise a “not connected” connection must be used at the input to avoid the input acting like an antenna, and prevent instability problems.

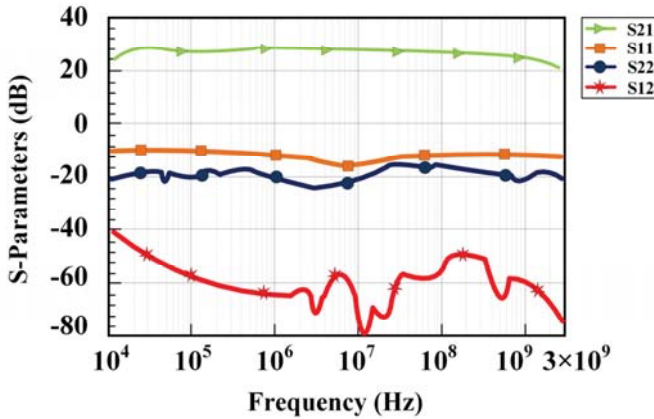


Fig. 5. Measured S-parameters of 0.18 μm TIA chip vs. frequency

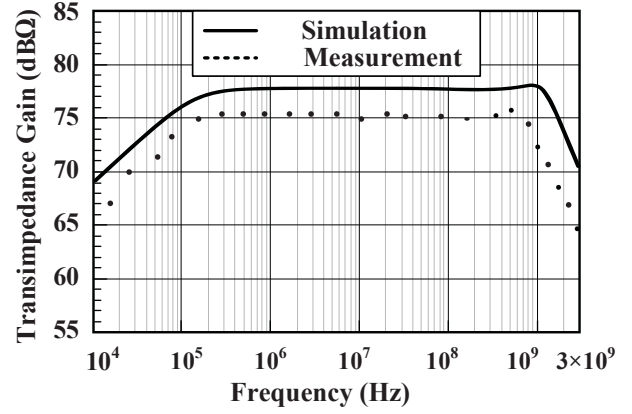


Fig. 6. Measured and post-layout simulated results of transimpedance gain for 0.18 μm TIA chip vs. frequency

Besides, the output noise voltage of the TIA might be lower than noise floor of the common spectrum analyzers, so extra amplification might be needed. In that case, after amplifying the output noise, the noise figure of the added stages must be taken into account.

Considering the above explanation, the final noise measurement result is shown in the Fig. 7 and is compared with the post layout simulation results.

The post-layout simulation results predicted input referred noise current of 1.64 pA/ $\sqrt{\text{Hz}}$ at 1 GHz and an average of 2.1 pA/ $\sqrt{\text{Hz}}$ within the bandwidth of the circuit. Measurement results shows an average input referred noise current of 3.18 pA/ $\sqrt{\text{Hz}}$ within the frequency bandwidth of the transimpedance amplifier. Slightly higher noise in the measurement results can be due to parasitic capacitances from PCB and measurement set-up which was added to input capacitance (C_{in}) and as predicted by Eq. 4, extra input capacitance results in higher input noise.

Regarding the proposed DC eliminating circuit, in test bench characterization, applying DC currents with the amplitudes of up to 10 μA , did not change the output voltage

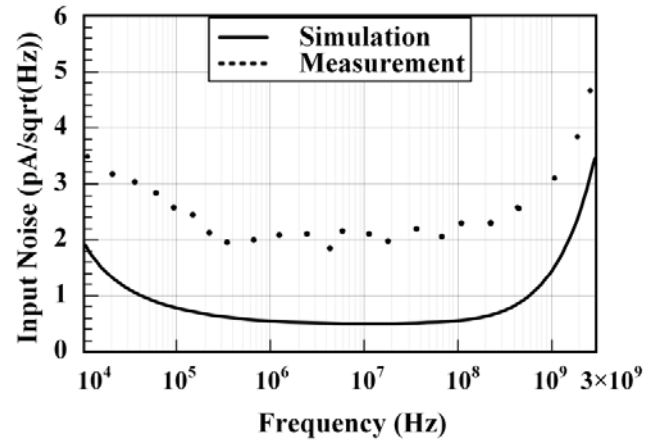


Fig. 7. Measured input referred current noise of the TIA and post-layout simulation results of 0.18 μm Technology vs. frequency.

TABLE I. Measurement and post-layout simulation results comparison with similar recent works.

	This Work Measured / Simulated	[9]	[7]	[11]	[12]	[13]	[15]	[16]	[17]
Technology	0.18 μm CMOS	0.6 μm CMOS	0.5 μm CMOS	0.25 μm CMOS	0.35 μm CMOS	0.18 μm CMOS	0.25 μm CMOS	40 nm CMOS	0.18 μm CMOS
TI Gain (dBΩ)	75.5 / 78	78	64	48	63.5	68	60.1	79.5	71.5
-3dB Bandwidth (GHz)	1.62 / 1.8	0.55	1.2	1.5	1.25	1.73	2.5	1.5	2.21
Power (mW)	26.3 / 26	30	20	72	40	NA	16	4.5	10
Avg. Input Noise Current (pA/$\sqrt{\text{Hz}}$)	3.18 / 2.1	4.5	15	9	16	3.3	*18.9	**8.85	*8.3

*Simulation results **Calculated by authors

of the TIA. In addition to that, the DC dark current mismatch of photodiodes in real experiment, which was close to 120 nA, did not cause saturation of TIA.

In table I, the measurement and post-layout simulation results are compared with results of other recent state of the art publications.

V. CONCLUSION

In this paper, a state of the art low-noise and high-gain transimpedance amplifier for 2.5 Gb/s optical communication circuits family was presented. A thorough design procedure for low noise and high gain TIA design was proposed, analytical noise optimization was performed, and optimum noise criteria were derived. A novel circuit level solution for dark current of input photodiodes were proposed and implemented. Measurement results of the chip, implemented in 0.18 μm TSMC CMOS technology, were shown and their conformity with analytical calculations and analysis were presented.

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